St. Pölten University of Applied Sciences

informatik & security

Modulhandbuch
FH-Bachelor Studiengang

IT Security

Stg KZ 0410

Department Informatik und Security

gemäß Fachhochschul-Akkreditierungsverordnung vom 28.05.2015
an das
Kollegium der FH St. Pölten

Antragsversion 1.4
St. Pölten, am 2021-06-29

1 Curriculum & Prüfungsordnung

1.1 Curriculum

1.1.1 Aufbau des Studienplanes

Der Studienplan ist nach § 3/2/1 FHStG aufgebaut. Er gewährleistet die Vielfalt wissenschaftlicher Lehrmeinungen und Methoden, berücksichtigt interdisziplinäre Lehrinhalte und vermittelt Fremdsprachenkompetenz. Die Ausbildung erstreckt sich von klassischen Fächern wie Mathematik und Programmieren über fachspezifische Fächer bis hin zu persönlichkeitsbildenden Fächern und Fremdsprachen.

Die fachspezifische Ausbildung besteht aus vier Schwerpunkten:

- Netzwerktechnik
- IT-Betrieb
- Sicherheitstechnologien
- Sicherheitsmanagement und Organisation

In jedem Schwerpunkt wird zuerst Basiswissen geschaffen, denn ohne fundiertes Grundlagenwissen ist eine Analyse und Bewertung von Sicherheitsrisiken sowie die Planung und Implementierung von Sicherheitskonzepten nicht möglich. Darauf bauen die – bei Fachhochschul-Studiengängen besonders ausgeprägten – praxisorientierten Anwendungen auf. In projektorientierten, teilweise fächerübergreifenden Lehrveranstaltungen werden die Studierenden an den Stand der Technik und zum selbstständigen wissenschaftlichen Arbeiten herangeführt.

Jeder Schwerpunkt besteht aus mehreren Modulen. In den Modulen werden Stoffgebiete zu thematisch und zeitlich abgerundeten, in sich abgeschlossenen und abprüfbaren Einheiten zusammengefasst. Zusätzlich werden in den Grundlagenmodulen andere didaktische Konzepte als in den Anwendungsmodulen verwendet, da sich Studierende in einer anderen Phase des Ausbildungsfortschritts befinden und die studentische Eigeninitiative differenzierter angesprochen werden kann. In diesem Ausbildungszyklus werden ein hoher Praxisbezug und die Förderung der Handlungskompetenz durch verstärkten Einsatz von Übungen, Fallstudien und Projekten hergestellt.

Im 4. Semester (Vollzeit-Studium) bzw. 5. Semester (berufsbegleitendes Studium) wird die erste Bachelorarbeit im Rahmen der Lehrveranstaltung „Wissenschaftliches Arbeiten" erstellt, wobei Gruppenaufgaben zulässig sind. Im Vollzeit-Studium wird im 5. Semester das verpflichtende Praktikum absolviert, die berufsbegleitenden Studierenden absolvieren das Praktikum während der gesamten Studiendauer verteilt. Im Vollzeit-Studium sind im 6. Semester ein fächerübergreifendes Projekt (berufsbegleitend: 6. Semester) und die zweite Bachelorarbeit vorgesehen (berufsbegleitend: 7. Semester). Mit der Projektarbeit soll das fächerübergreifende, selbstständige Arbeiten gefördert und die Studierenden sollen auf die Aufgaben, die sie in der Wirtschaft erwarten, vorbereitet werden.

1.1.2 Grafische Darstellung der Module

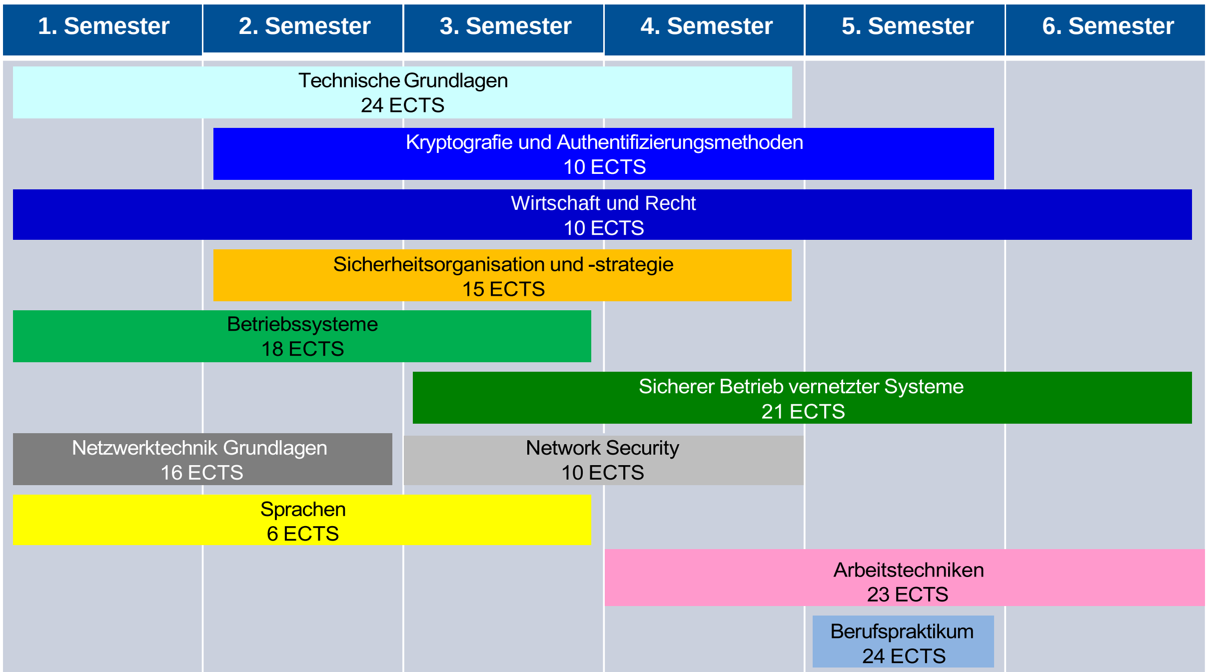

Abbildung 1: Grafische Darstellung der Module VZ

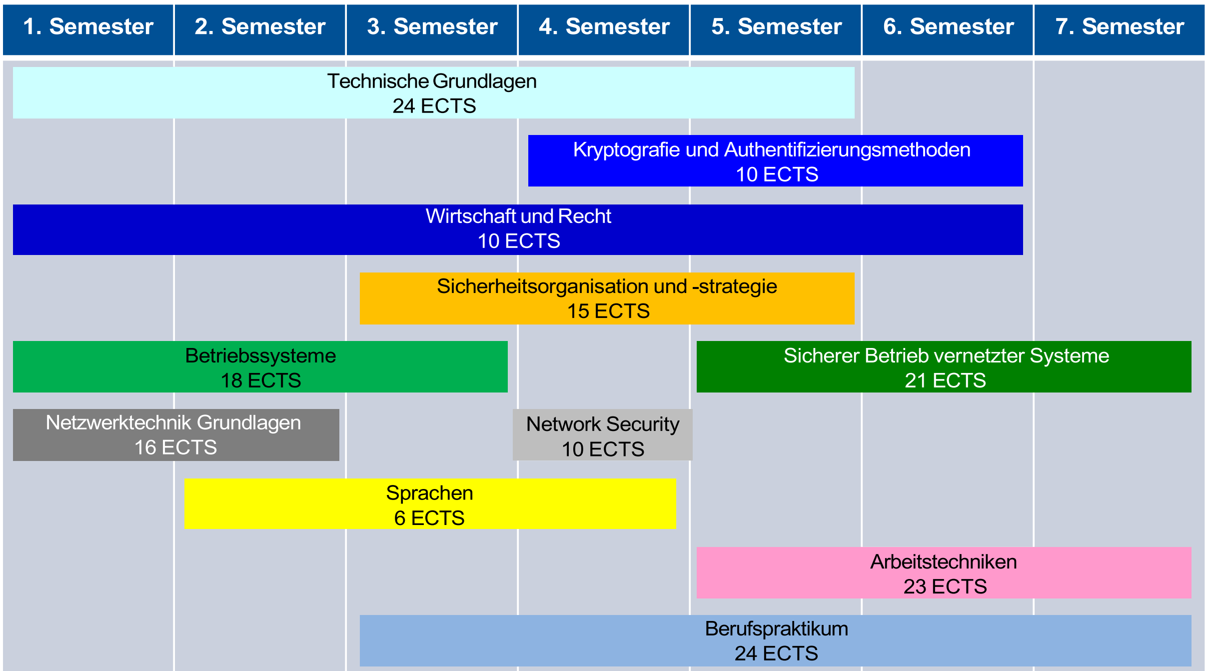

Abbildung 2: Grafische Darstellung der Module BB

1.2 Curriculum und Modulbeschreibungen

1.2.1 Anlage: Curriculumsdaten

	VZ	BB	Allfälliger Kommentar
Erstes Studienjahr	2011/12	2013/14	-
Regelstudiendauer	6 Semester	7 Semester	-
Pflicht-SWS	111	111	-
LV-Wochen pro Semester	15	15	Basis für alle weiteren Kalkulationen
Pflicht-LVS	900	900	-
Pflicht-ECTS	180	180	-
			-
WS Beginn	KW38	KW37	-
WS Ende	KW5	KW5	-
SS Beginn	KW8	KW8	-
SS Ende	KW37	KW37	
WS Wochen	ca. 18	ca. 21	variiert von Jahr zu Jahr, je nach Kalender und Ferieneinteilungen
SS Wochen	ca. 18	ca. 21	
	Deutsch	Deutsch	
Verpflichtendes Auslandssemester	-	-	nicht vorgesehen, individuelles Auslandssemester aber gefördert
Unterrichtssprache	Deutsch	Deutsch	einzelne, ausgewählte Lehrveranstaltungen können auch in englischer Sprache abgehalten werden
Berufspraktikum	5. Semester	3.-7. Semester	24 ECTS (600 Stunden)
Resultiert aus Überführung des Studiengangs		-	

1.2.2 Darstellung des Curriculums Vollzeit-Studium

1. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
ITB101	Windows-Systemadministration	ILV	3	2	5	75	Betriebssysteme	5
NWT101	Einführung Netzwerke und verteilte Systeme	ILV	5	2	8	120	Netzwerke -Grundlagen	9
SPR101	Englisch 1	ILV	2	1	4	60	Sprachen	2
TECH101	Grundzüge der diskreten Mathematik	ILV	5	1	5	75	Technische Grundlagen	7
TECH103	Programmieren 1	ILV	4	2	6	90	Technische Grundlagen	5
SIMAN101	Teamtraining	ILV	2	2	4	60	Wirtschaft und Recht	2
Summenzeile			**21**		**32**	**480**		**30**
LVS = SummeSWS*LV-Wochen			**315**					

2. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
ITB103	Serverbetrieb und Hardening Windows	ILV	2	2	3	45	Betriebssysteme	3
ITB102	Unix Systemadministration	ILV	3	2	5	75	Betriebssysteme	4
SITEC101	Grundlagen der Kryptografie	ILV	5	2	8	120	Kryptografie und Authentifizierungs-methoden	5
NWT102	Aktive Komponenten und Applikationsprotokolle	ILV	6	2	10	150	Netzwerke- Grundlagen	7
SIMAN201	Sicherheitsmanage-ment 1- Basics und Systeme	ILV	2	2	3	45	Sicherheitsorganisation und -Strategie	3
SPR102	Englisch 2	ILV	2	2	4	60	Sprachen	2
TECH105	Programmieren 2	ILV	5	2	6	90	Technische Grundlagen	6
Summenzeile			**25**		**41**	**615**		**30**
LVS = SummeSWS*LV-Wochen			**375**					

3. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
ITB203	Web und Applikationssicherheit	ILV	3	2	5	75	Sicherer Betrieb vernetzter Systeme	5
ITB104	Serverbetrieb und Hardening Unix	ILV	4	2	6	90	Betriebssysteme	6
NWT202	Netzwerk Sicherheitskompo-nenten	LB	2	2	4	60	Network Security	3
NWT201	Netzwerk Sicherheitskompo-nenten	VO	1	1	1	15	Network Security	2
SIMAN105	IT Prozesse nach ITIL	ILV	2	2	3	45	Sicherheitsorganisation und -Strategie	3
SIMAN202	Sicherheitsmanage-ment 2- Code of Practise	ILV	4	2	5	75	Sicherheitsorganisation und -Strategie	4
SPR103	Englisch 3	ILV	2	2	4	60	Sprachen	2
TECH107	Programmieren 3	ILV	4	2	6	90	Technische Grundlagen	5
Summenzeile			**22**		**34**	**510**		**30**
LVS = SummeSWS*LV-Wochen			**330**					

4. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
ARB101	Methoden des wissenschaftlichen Arbeitens	ILV	2	5	7	105	Arbeitstechniken	4
SITEC104	Digitale Forensik	ILV	2	2	3	45	Kryptografie und Authentifizierungsmethod en	3
NWT203	Secure Networks	ILV	4	2	6	90	Network Security	5
ITB201	Datenbanksysteme	ILV	3	2	4	60	Sicherer Betrieb vernetzter Systeme	4
TECH108	Programmieren 4	ILV	3	2	5	75	Technische Grundlagen	5
ITB202	Essential Hacking Tools and Techniques	ILV	3	2	5	75	Sicherer Betrieb vernetzter Systeme	4
SIMAN203	Sicherheitsmanage-ment 3 - Risiko und Notfall	ILV	4	2	6	90	Sicherheitsorganisation und -Strategie	5

Summenzeile		**21**		**36**	**540**		**30**
LVS = SummeSWS*LV-Wochen		**315**					

5. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
BP106	Betreuungsseminar Praktikum	SE	2	2	4	60	Berufspraktikum	4
BP101	Praktikum	BP	0	2	0	0	Berufspraktikum	20
SITEC103	Biometrie	ILV	2	1	2	30	Kryptografie und Authentifizierungsmethoden	2
SIMAN104	Sicherheitsmanagement 4- Outsourcing	ILV	3	2	4	60	Wirtschaft und Recht	4
Summenzeile			**7**		**10**	**150**		**30**
LVS = SummeSWS*LV-Wochen			**105**					

6. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
ARB102	Bachelorseminar mit Bachelorarbeit	SE	3	7	12	180	Arbeitstechniken	12
ARB108	Wahlpflichtfach	PR	3	3	8	120	Arbeitstechniken	7
ITB205	Business Continuity und Disaster Recovery	UE	2	2	4	60	Sicherer Betrieb vernetzter Systeme	2
ITB204	Security Audits & Penetration Testing	ILV	4	2	6	90	Sicherer Betrieb vernetzter Systeme	6
SIMAN106	Rechtliche Grundlagen	VO	3	1	3	45	Wirtschaft und Recht	3
Summenzeile			**15**		**33**	**495**		**30**
LVS = SummeSWS*LV-Wochen			**225**					

Summe über alle Semester							
Summenzeile		**111**		**184**	**2760**		**180**
LVS = SummeSWS*LV-Wochen		**1665**					

1.2.3 Darstellung des Curriculums berufsbegleitendes Studium

1. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
ITB101	Windows System-administration	ILV	3	1	3	45	Betriebssysteme	5
NWT101	Einführung Netzwerke und verteilte Systeme	ILV	5	1	5	90	Netzwerke-Grundlagen	9
TECH101	Grundzüge der diskreten Mathematik	ILV	5	1	5	75	Technische Grundlagen	7
SIMAN 101	Teamtraining	ILV	2	1	2	30	Wirtschaft und Recht	2
Summenzeile			**15**		**15**	**225**		**23**
LVS = SummeSWS*LV-Wochen			**225**					

2. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
TECH103	Programmieren 1	ILV	4	1	4	60	Technische Grundlagen	5
ITB102	Unix Systemadministration	ILV	3	1	3	45	Betriebssysteme	4
SITEC101	Grundlagen der Kryptografie	ILV	5	1	5	75	Kryptografie und Authentifizierungs-methoden	5
NWT102	Aktive Komponenten und Applikationsprotokolle	ILV	6	1	7	105	Netzwerke-Grundlagen	7
SPR101	Englisch 1	ILV	2	1	2	30	Sprachen	2
Summenzeile			**20**		**20**	**300**		**23**
LVS = SummeSWS*LV-Wochen			**300**					

3. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
BP101	Praktikum	BP	0	1	0	0	Berufspraktikum	4
ITB103	Serverbetrieb und Hardening Windows	ILV	2	1	2	30	Betriebssysteme	3
SPR10 2	Englisch 2	ILV	2	1	2	30	Sprachen	2
SIMAN 102	Teamtraining 2	ILV	1	1	1	15	Wirtschaft und Recht	1
TECH1 05	Programmieren 2	ILV	5	1	4	60	Technische Grundlagen	6
SIMAN 201	Sicherheitsmanagement 1- Basics und Systeme	ILV	2	1	2	30	Sicherheitsorganisation und –Strategie	3
ITB104	Serverbetrieb und Hardening Unix	ILV	4	1	4	60	Betriebssysteme	6
SIMAN 105	IT Prozesse nach ITIL	ILV	2	1	2	30	Sicherheitsorganisation und -Strategie	3
Summenzeile			**17**		**17**	**225**		**27**
LVS = SummeSWS*LV-Wochen			**225**					

4. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
BP101	Praktikum	BP	0	1	0	0	Berufspraktikum	4
NWT20 2	Netzwerk Sicherheitskomponenten	LB	2	1	1	15	Network Security	3
NWT20 1	Netzwerk Sicherheitskomponenten	VO	1	1	1	15	Network Security	2
TECH1 07	Programmieren 3	ILV	4	1	4	60	Technische Grundlagen	5
SIMAN 202	Sicherheitsmanagement 2- Codes of Practise	ILV	4	1	4	60	Sicherheitsorganisation und –Strategie	4
SPR10 3	Englisch 3	ILV	2	1	2	30	Sprachen	2
SITEC1 04	Digitale Forensik	ILV	2	1	2	30	Digitale Forensik	3
NWT20 3	Secure Networks	ILV	4	1	4	60	Secure Networks	5
Summenzeile			**19**		**19**	**285**		**28**
LVS = SummeSWS*LV-Wochen			**285**					

5. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
BP101	Praktikum	BP	0	1	0	0	Berufspraktikum	4
TECH108	Programmieren 4	ILV	3	1	3	45	Technische Grundlagen	5
ITB202	Essential Hacking Tools and Techniques	ILV	3	1	3	45	Sicherer Betrieb vernetzter Systeme	4
ITB201	Datenbanksysteme	ILV	3	1	3	45	Sicherer Betrieb vernetzter Systeme	4
SIMAN 203	Sicherheitsmanagement 3- Risiko- und Notfall	ILV	4	1	4	60	Sicherheitsorganisation und –Strategie	5
ARB101	Methoden des wissenschaftlichen Arbeitens	ILV	2	1	4	60	Arbeitstechniken	4
Summenzeile			**15**		**15**	**225**		**26**
LVS = SummeSWS*LV-Wochen			**225**					

6. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	*ECTS*
BP101	Praktikum	BP	0	1	0	0	Berufspraktikum	4
ITB203	Web- und Applikations-sicherheit	ILV	3	1	3	45	Sicherer Betrieb vernetzter Systeme	5
SITEC103	Biometrie	ILV	2	1	2	30	Kryptografie und Authentifizierungs-methoden	2
SIMAN104	Sicherheitsmanage-ment 4 - Outsourcing	ILV	3	1	3	45	Wirtschaft und Recht	4
ARB108	Fächerübergreifende Projektarbeit	PR	3	2	4	60	Arbeitstechniken	7
SIMAN106	Rechtliche Grundlagen	VO	3	1	3	45	Wirtschaft und Recht	3
Summenzeile			**14**		**15**	**225**		**25**
LVS = SummeSWS*LV-Wochen			**210**					

7. Semester

LV-Nr	LV-Bezeichnung	LV-Typ	SWS	Anzahl Gruppen	ASWS	ALVS	Modul	ECTS
BP101	Praktikum	BP	0	1	0	0	Berufspraktikum	4
BP106	Betreuungsseminar Praktikum	SE	2	1	2	30	Berufspraktikum	4
ITB204	Security Audits & Penetration Testing	ILV	4	1	4	60	Sicherer Betrieb vernetzter Systeme	6
ARB102	Bachelorseminar und Bachelorarbeit 2	SE	3	2	5	75	Arbeitstechniken	12
ITB205	Business Continuity und Disaster Recovery	UE	2	1	2	30	Sicherer Betrieb vernetzter Systeme	2
Summenzeile			**11**		**13**	**195**		**28**
LVS = SummeSWS*LV-Wochen			**165**					

Summe über alle Semester								
Summenzeile			**111**		**114**	**1710**		**180**
LVS = SummeSWS*LV-Wochen			**1665**					

1.2.4 Modularisierung des Curriculums

Technical Basics

Modulnummer:	**Modultitel:**	**Umfang:**
TECH1	**Technical Basics**	**27 ECTS**
Studiengang	Bachelor IT Security	
Lage im Curriculum	1. bis 4. Semester (VZ, BB)	
Zuordnung zu den Teilgebieten	Technical Basics	
Niveaustufe	Grundlagen	
Vorkenntnisse	Keine	
Geblockt	Die einzelnen LVs werden geblockt abgehalten	
Kreis d. TeilnehmerInnen	Studierende des Studienganges IT Security	
Beitrag zu nachfolgenden Modulen	Betriebssysteme; Netzwerke Grundlagen; Network-Security; Betrieb vernetzter Systeme; Kryptografie und Authentifizierung	
Literaturempfehlungen	[1] A. Willms, C Programmieren lernen. Addison Wesley [2] P. Prinz und U. Kirch-Prinz, C. Kurz und gut. O'Reilly Media. [3] D. Ratz, J. Scheffler, D. Seese, und J. Wiesenberger, Grundkurs Programmieren in Java. Hanser. [4] G. Krüger und T. Stark, Handbuch der Java-Programmierung. Addison-Wesley Longman Verlag. [5] S. Oaks, Java Security. O'Reilly Media. [6] D. Hachenberger, Mathematik für Informatiker. Pearson Education. [7] G. Teschl und S. Teschl, Mathematik für Informatiker. Springer. [8] R. Klima, Programmieren in C. Springer. [9] T. Kheis, Einstieg in Python. Galileo. [10] E. Ramalho, Fluent Python. O`Reilley [11] Java ist auch eine Insel. Ullenboom, Rheinwerk.	
Learning Outcomes	▪ The students improve their logical thinking by getting acquainted with Boolean algebra and number theory. ▪ They can put into practice selected chapters of combinatorics, information theory, and number theory, in particular in cryptography and statistics algorithms. ▪ The students develop, code, and test programmes in the programming language C. ▪ They analyse security-critical aspects of the systems programming language C and use pointers correctly. ▪ Upon completion of the programming training, they develop programmes according to an object-oriented model and write applications in an OO programming language such as Java or Python that can be used for security analyses. ▪ They can describe the major aspects of developing secure and stable software and apply them to their own programmes. ▪ They know the most important technologies for implementing web and enterprise applications in Java or C# and, in part, they are able to implement them autonomously in projects as well as analyse and evaluate them.	
Titel der Lehrveranstaltung	Basics of Discrete Mathematics	
Umfang	3 SWS (VO) + 2 SWS (UE), 4 ECTS (VO) + 3 ECTS (UE)	
Lage im Curriculum	1. Semester (VZ+BB)	

Lehr- und Lernformen	Theorie- und Praxiseinheiten wechseln sich ab: Theorie, Verfahren und Algorithmen erklären und analysieren, Beispiele rechnen, ein weiteres Verfahren / Algorithmus selbst präsentieren
Prüfungsmodalitäten	Beispiele rechnen
Learning Outcomes	▪ The students have improved their logical thinking by learning Boolean algebra and number theory. ▪ They can explain selected chapters of combinatorics, information theory, IT mathematics, and number theory and work out practical examples. ▪ They can also explain and use the time-efficient implementations of cryptography algorithms (modular multiplication and potentiation), practical algorithms (such as prime number test, generation of random figures, CRC calculation, and data compression), and selected chapters of statistics.
Course Contents	Quantities of numbers, number systems, logics and volumes, vector spaces and matrices, combinatorics, statistics, relations/functions, addition, subtraction, multiplication and division in the dual system, number representation plus conversion, floating-point representation, Boolean algebra (gate, half-adder, serial adder, HW multiplier, semiconductor memory), coding, block codes, Hamming distance, recognition and correction of errors, information theory (information content, entropy, redundancy, redundancy reduction), compression, source trees, Huffmann code, number theory (congruencies, basic arithmetical operations in modular arithmetic, prime numbers, Fermat's theorem, Rabin-Miller prime number test, potentiation of large numbers, Montgomery procedure, Chinese remainder theorem, groups, rings, bodies, polynomials in the body, Galois field, CRC calculation, Euler's totient function, Euler's theorem, quadratic residue, square root calculation according to Rabin and AMM procedure, discrete logarithm, primitive root, random numbers, secret sharing, Shamir secret sharing, elliptic curves (point addition, point multiplication), students' own development and presentation.
Titel der Lehrveranstaltung	Programming 1
Umfang	4 SWS, 5 ECTS
Lage im Curriculum	1. Semester (VZ) bzw. 2. Semester (BB)
Lehr- und Lernformen	ILV
Prüfungsmodalitäten	LV-immanenter Prüfungscharakter im Übungsteil, praktische Programmierarbeit als Abschlussprüfung
Learning Outcomes	▪ Upon completion of programming 1, students can formulate algorithms in an (object-oriented) programming language. They implement the basic programming concepts in their own applications. ▪ They draft, implement, and test their own small software projects using the programming language Python. ▪ The students use Python as a typical tool for implementing smaller software components to support them in their daily responsibilities in the software environment (cryptography, network technology, system administration). ▪ They choose the right algorithms for programming tasks. ▪ They integrate algorithms into their own programme. ▪ Upon completion of the course, students can formulate tasks as structured algorithms and implement and test them as Python programmes using the available control structures.

Course Contents	▪ Syntax of the language Python ▪ Types, variables, and constants ▪ Algorithm design ▪ Good overview of the start-of-the-art technologies used and the principles of OO programming; students can use Python as a tool to automatise the practical work of a security expert. ▪ Classes, instances, methods and attributes, references, inheritance and interfaces, polymorphism, abstract classes, generics; practical examples using the OO language Python.
Titel der Lehrveranstaltung	Programming 2
Umfang	5 SWS, 6 ECTS
Lage im Curriculum	2. Semester (VZ) bzw. 3. Semester (BB)
Lehr- und Lernformen	ILV
Prüfungsmodalitäten	LV-immanenter Prüfungscharakter im Übungsteil, praktische Programmierarbeit als Abschlussprüfung
Learning Outcomes	After completing this course, the students can independently put advanced techniques in the programming language C to use in their own programmes: ▪ They can define, combine, and apply data structures (in the programming language C) for the storage and manipulation of information in such a way that no security gaps arise. ▪ They solve algorithms with the help of functions and, in doing so, apply the correct form of parameters (value or reference, constant or modifiable). ▪ The students are able to construct secure and stable programmes using pointers. ▪ They use the most common recursive data structures in their own programmes. ▪ They programme recursive algorithms. ▪ They use macros available in C and formulate their own. ▪ The students define function pointers and use them as parameter or in structures and vectors. ▪ They divide larger programming projects into modules and formulate them in Build environments (e.g., *make*).
Course Contents	▪ Syntax of the language C ▪ Pointers & vectors ▪ Functions and parameters ▪ Management of the development environment for implementing and testing ▪ Use of structs and IO streams ▪ Function pointers and their application ▪ C preprocessor & macros ▪ Recursive data structures, recursive programming and the standard algorithms implemented with it ▪ Module building
Titel der Lehrveranstaltung	Programming 3
Umfang	4 SWS, 5 ECTS
Lage im Curriculum	3. Semester (VZ) bzw. 4. Semester (BB)
Lehr- und Lernformen	ILV
Prüfungsmodalitäten	LV-immanenter Prüfungscharakter im Übungsteil, praktische Programmierarbeit als Abschlussprüfung
Learning Outcomes	▪ Having completed Programming 3, the students describe common object-oriented programming concepts in modern

	"Managed Environments" such as Java or .Net (classes, instances, references, derivation, polymorphism, interfaces, access methods, etc.). ▪ They independently design, implement, and test smaller software projects in Java (or C#). ▪ They students have a good overview of the techniques used and the libraries available in Java (or .Net) and can discuss the architecture of software products with developers.

Course Contents	While this course does not turn students into professional software developers, they still gain a good overview of the latest technologies and principles of OO programming (classes, instances, methods and attributes, references, inheritance and interfaces, polymorphism, abstract classes, generics). They design, implement, and test smaller applications on their own in a modern programming language such as Java or C#.
Titel der Lehrveranstaltung	Programming 4
Umfang	3 SWS, 5 ECTS
Lage im Curriculum	4. Semester (VZ) bzw. 5. Semester (BB)
Lehr- und Lernformen	ILV
Prüfungsmodalitäten	LV-immanenter Prüfungscharakter im Übungsteil, praktische Programmierarbeit als Abschlussprüfung
Learning Outcomes	▪ The students focus on security aspects in (primarily object-oriented) software development and know typical danger points and how to avoid them. ▪ They discuss and verify security aspects in OO software development. ▪ They have a good overview of current techniques in the development of web and enterprise applications.
Course Contents	- One key focus is on typical security aspects in software development such as the top 10 of OWSAP that are tested and experienced in practical exercises. - Moreover, the students are familiar with common techniques for the implementation of web and enterprise applications and they acquire the necessary basic understanding for carrying out security-related tests and analyses in (larger) software projects.

Cryptography and Authentication Methods

Modulnummer:	**Modultitel:**	**Umfang:**
SITEC 1	**Cryptography and Authentication Methods**	**10 ECTS**
Studiengang	Bachelor IT Security	
Lage im Curriculum	2., 4. und 5. Semester (VZ) bzw. 2., 4. und 6. Semester (BB)	
Zuordnung zu den Teilgebieten	Sicherheitstechnologien	
Niveaustufe	Basis	
Vorkenntnisse	Zahlentheorie	
Geblockt	Die einzelnen LVs werden geblockt abgehalten	
Kreis d. TeilnehmerInnen	Studierende des Studienganges IT Security	
Beitrag zu nachfolgenden Modulen	Voraussetzung für die Module NWT2 und ITB2	
Literaturempfehlungen	[1] B. Schneier, „Angewandte Kryptografie“, Pearson Education [2] A. Menezes et al., “Handbook of Applied Cryptography”, CRC Press [3] Daemen et al, „The Design of Rijndael“. Berlin. 2002 [4] N. Koblitz, „Algebraic Aspects of Cryptography“, Springer, 1999 [5] A. Werner, „Elliptische Kurven in der Kryptographie“, Springer, 2002 [6] D. Bernstein et al., „Post-Quantum Cryptography“, Springer, 2009. [7] J. Junqing et al., "Key generation from wireless channels: A review." IEEE Access 4, 614-626, 2016 [8] A. Beutelspacher et al, „Moderne Verfahren der Kryptographie“,Vieweg [9] O. Goldreich et al., “Proofs that Yield Nothing But Their Validity”, Journal of the ACM, Volume 38, S. 690ff, 1991 [10] M. Behrens, R. Roth, „Biometrische Identifikation - Grundlagen, Verfahren, Perspektiven“, Verlag Vieweg, 2001 [11] J. Ashbourn, „Practical Biometrics“, Springer, 2004 [12] Bolle et al., „Guide to Biometrics“, Springer, 2003 [13] E. Casey, „Handbook of digital forensics and investigation“, Academic Press, 2009 [14] E. Casey, „Digital evidence and computer crime: Forensic science, computers, and the internet“, Academic press, 2011. [15] H. Carvey, „Windows forensic analysis“, Syngress, 2009 [16] S. Garfinkel, „Anti-forensics: Techniques, detection and countermeasures“, 2nd International Conference on i-Warfare and Security, 2007 [17] M.H. Ligh et al., „The art of memory forensics: Detecting malware and threats in Windows, Linux, and Mac memory“, John Wiley & Sons, 2014	
Learning Outcomes	▪ The students are familiar with common processes and algorithms of cryptography and key management and can apply them and calculate practical examples. ▪ They understand the IT protection objectives integrity, confidentiality and authenticity, and know which IT security problems can be solved with the help of cryptography. ▪ Having acquired the metadisciplinary competence “Presenting”, they can plan a lecture and create a presentation within the predefined framework conditions. ▪ In addition, they are familiar with user authentication based on password and biometrical methods and can explain and use the most common methods and procedures. ▪ Having acquired the metadisciplinary competence “Ethics”, they can explain ethical questions of data protection and identify	

	ethical problems in the security field. ▪ In the field of digital forensics, the students are able to explain and put into practice common analysis methods. The results can be presented in the form of a report to interest groups in the field of civil or criminal law.

Titel der Lehrveranstaltung	Basics of Cryptography
Umfang	5 SWS, 5 ECTS
Lage im Curriculum	2. Semester (VZ+BB)
Lehr- und Lernformen	Theorie- und Praxiseinheiten wechseln sich ab: Verfahren / Algorithmen erklären, herleiten, analysieren und bewerten, Beispiele rechnen; Planung, Erstellung und Halten einer Präsentation
Prüfungsmodalitäten	Vorlesungsnote: Multiple-Choice Prüfung. Übungsnote: Beispiele rechnen und Präsentation, alle Teile müssen positiv sein
Learning Outcomes	▪ The students are familiar with common processes and algorithms of symmetrical and asymmetrical cryptography and key management and can apply them and calculate practical examples. ▪ They understand the IT protection objectives integrity, confidentiality and authenticity, and know which IT security problems can be solved with the help of cryptography. They are also able to select the optimal procedure or algorithm. ▪ Having acquired the metadisciplinary competence “Presenting”, they can plan and create the structure of a lecture and give a presentation within the predefined framework conditions (time, target audience, etc.).
Course Contents	Historical overview, security problems that can be solved with the help of cryptography, encryption modes: ECB, CBC, CTR, GCM, XTS, OFB, CFB; encryption procedures: tripleDES, AES, RSA, ElGamal with “discrete logarithm“ and “elliptic curves”; hash functions (SHA-1/-2/-3), MAC (HMAC, CBC-MAC, GCM-MAC); Digital signature: basics, procedures (RSA, DSA, ECDSA), PKI, blind signature (Chaum’s scheme); Cryptographic authentication: challenge response procedure, zero-knowledge procedure (Fiat-Shamir, Guillou-Quisquater); Key management: generation, storage, erasure, distribution, DH and ECDH procedure, key derivation, DDA procedure; multiparty computations (3 procedures); quantum cryptography; cryptography standards (ISO, PKCS#), politics and cryptography, cryptography software; cryptographic security, side channel attacks; post-quantum cryptography; key generation by measuring radio channel features; next to the lecture practical approach to the course contents. Metadisciplinary competence “Presenting”: planning and creating the structure of a lecture, giving presentations within the predefined framework conditions (time, target audience, etc.), analysing presentations of other students, knowing the means of expression of body language and the voice, giving constructive feedback.

Titel der Lehrveranstaltung	Digital Forensics
Umfang	2 SWS, 3 ECTS
Lage im Curriculum	4. Semester (VZ+BB)

Lehr- und Lernformen	Vortrag, Übungsaufgaben
Prüfungsmodalitäten	Abgabe von Laborprotokollen (80%), Abgabegespräche mit theoretischer Komponente (20%)
Learning Outcomes	▪ The students are familiar with the basics of IT forensics and can put them to practical use. ▪ They know the organisational and legal basics and have basic knowledge of the structure and analysis of various data systems; they can take care of the acquisition or live analysis of information technology systems and are skilled in the digital forensics of smartphones and tablets and know common anti-forensic techniques. ▪ Students have deepened their knowledge in IT forensics by writing a forensics report based on a training case. ▪ They can write a forensic report adhering to formal criteria and the basic principles of digital forensics.
Course Contents	Organisational aspects and best practices of digital forensics, legal framework conditions, theoretical and practical acquisition of an image disk (data carrier, RAM) via software and hardware, memory forensics for Windows & Linux with the Tool Volatility, properties of smartphone forensics (iOS and Android), data carrier forensics (NTFS image analysis with autopsy, theoretical basics of further data systems), practical analysis of smartphones and tables with various tools, anti-forensic methods (data hiding, falsification) and basics of steganography. Content and structure of a forensic report, writing a report.
Titel der Lehrveranstaltung	Biometry
Umfang	2 SWS, 2 ECTS
Lage im Curriculum	5. Semester (VZ) bzw. 6. Semester (BB)
Lehr- und Lernformen	Theorie- und Praxiseinheiten wechseln sich ab
Prüfungsmodalitäten	Übungen während der ILV. Schriftliche Prüfung. Ausarbeitung in Gruppenarbeit. Alle Teile müssen positiv sein. Gewichtung in %: 60% Prüfung, 40% Beispiele
Learning Outcomes	▪ The students are familiar with user authentication based on passwords and biometrical methods and know the most important methods and procedures. They can explain and apply the most common methods and procedures. ▪ Having acquired the metadisciplinary competence "Ethics", the students can explain ethical topics pertaining to data protection and identify ethical problems in the field of security.

Course Contents	Terminology, simple passwords, password guidelines, password memory aids, password algorithms; Biometrical methods: advantages, disadvantages, FRR, FAR, FER, conditioned/random/inherited features, fingerprint, analysis of handwriting, iris scan, facial recognition, vein recognition, voice, hand geometry, keyboard input, mouse recognition; Verification vs. identification, problems caused by biometry; Introduction to identity management; Practical example; literature references, next to the lecture the course content will be put to practical use Metadisciplinary competence "Ethics": Explaining ethical data protection issues, identifying ethical problems in the field of security.

Economics and Law

Modulnummer:	Modultitel:	Umfang:
SIMAN1	**Economics and Law**	**10 ECTS**
Studiengang	Bachelor IT Security	
Lage im Curriculum	1., 5.und 6. Semester (VZ+ BB)	
Zuordnung zu den Teilgebieten	Sicherheitsmanagement	
Niveaustufe	Grundlagen	
Vorkenntnisse	Keine	
Geblockt	Die einzelnen LVs werden geblockt abgehalten	
Kreis d. TeilnehmerInnen	Studierende des Studienganges IT Security	
Beitrag zu nachfolgenden Modulen	Sicherheitsorganisation und -strategie	
Literaturempfehlungen	[1] European Network and information Security Agency (ENISA), “A Users’ Guide: How to raise information security awareness.” [2] K. Kistner und M. Steven, Betriebswirtschaftslehre im Grundstudium. Physica-Verlag. [3] K. Mitnick und W. Simon, Die Kunst der Täuschung - Risikofaktor Mensch. mitp. [4] B. Funk, Einführung in das österreichische Verfassungsrecht. Leykam. [5] G. Wielinger, Einführung in das österreichische Verwaltungsverfahrensrecht. Leykam. [6] K. Lechner, A. Egger, und R. Schauer, Einführung in die Allgemeine Betriebswirtschaftslehre. Linde. [7] G. Wöhe und U. Döring, Einführung in die Allgemeine Betriebswirtschaftslehre. Vahlen. [8] C. Grohmann-Steiger, W. Scheider, und E. Eberhartinger, Einführung in die Buchhaltung im Selbststudium. Facultas Universitätsverlag. [9] IT-Governance Institute, “Governance of Outsourcing,” IT Governance Institute. [10] P. Kotler, G. Armstrong, J. Saunders, und V. Wong, Grundlagen des Marketing. Pearson. [11] Bundesamt für Sicherheit in der Informationstechnik (BSI), “IT-Grundschutz-Standards.” [12] W. Kemmetmüller und S. Bogensberger, Kostenrechnung. Facultas Universitätsverlag. [13] M. Bruhn, Marketing. Grundlagen für Studium und Praxis. Gabler Verlag. [14] H. Hermes und G. Schwarz, Outsourcing: Chancen und Risiken, Erfolgsfaktoren, rechtssichere Umsetzung. Haufe-Lexware. [15] D. Pokoyski und M. Helisch, Security Awareness: Neue Wege zur erfolgreichen Mitarbeiter-Sensibilisierung. Vieweg+Teubner.	
Learning Outcomes	▪ The students feel competent integrating legal topics in their professional activities. ▪ They are also able to use relevant communication techniques to lead meetings, audits, or conflict situations.	

Titel der Lehrveranstaltung	Team Training 2
Umfang	1 SWS, 1 ECTS
Lage im Curriculum	2. Semester (VZ) bzw.3. Semester (BB)
Lehr- und Lernformen	Theorie-Input, Gruppenarbeiten, Einzelcoachings
Prüfungsmodalitäten	Auswertung der Teamarbeiten
Learning Outcomes	▪ The students can use exam preparation techniques.
Course Contents	Continuation of topics from Team Training 1 Mental training Teamwork skills

Titel der Lehrveranstaltung	Legal Basics
Umfang	3 SWS, 3 ECTS
Lage im Curriculum	6. Semester (VZ + BB)
Lehr- und Lernformen	Gruppenarbeiten in Projektform, Quiz, Vortrag, Stationenbetrieb
Prüfungsmodalitäten	Schriftliche Prüfung
Learning Outcomes	The students are able to integrate legal topics into their professional activities. They can understand, interpret, and use legal texts. They are informed of national and international legislation. Metadisciplinary competence Ethics: ▪ The students can discuss the difference between personality rights on the Internet and physical rights. (2) ▪ The students can explain the boundaries of data ethics. (2) ▪ The students can explain the terms “Data Protection” and “Data Misuse”. (2) ▪ The students can explain issues pertaining to legal ethics.
Course Contents	Introduction to the basic principles of law (structure of the legal system / difference between public and private law) Basics of personal law Basics of property law (possession, ownership) eCommerce law Basics of data protection Special characteristics of telecommunications law

Titel der Lehrveranstaltung	Security Management 4 – Outsourcing
Umfang	3 SWS, 4 ECTS
Lage im Curriculum	5. Semester (VZ) bzw. 6. Semester (BB)
Lehr- und Lernformen	Lehrziele werden durch Vorlesungen und Übungsteile (z.B.: Ausarbeitung einer Security Awareness Schulung, Planung eines Social Engineering Angriffs …) durchgeführt. In Zusammenarbeit mit einem industriellen Partner wird ein Workshop durchgeführt um eine realistisches Beispiel einer Vorstandspräsentation zum Thema Outsourcing der IT durchzuführen.

Prüfungsmodalitäten	2 Präsentationen + Ausarbeitung; Test
Learning Outcomes	▪ The students are familiar with the most important basic terms of outsourcing and of sensitising employees for information security. ▪ Having completed this course, the students can assess outsourcing projects as well as plan and carry out awareness trainings. ▪ They are able to summarise complex issues in clear management summaries.
Course Contents	Outsourcing Lifecycle / ISO 38500 ENISA Awareness Guide Interdisciplinary Competence for Text Types 2: Structure of management summaries, pyramid principle (Minto)

Security Organisation and Strategy

Modulnummer:	**Modultitel:**	**Umfang:**
SIMAN2	**Security Organisation and Strategy**	**15 ECTS**
Studiengang	Bachelor IT Security	
Lage im Curriculum	2., 3. und 4. Semester (VZ) bzw. 3., 4. und 5. Semester (BB)	
Zuordnung zu den Teilgebieten	Sicherheitsmanagement und Organisation	
Niveaustufe	Grundlagen	
Vorkenntnisse	Keine	
Geblockt	Die einzelnen LVs werden geblockt abgehalten	
Kreis d. TeilnehmerInnen	Studierende des Studienganges IT Security	
Beitrag zu nachfolgenden Modulen	Grundlegende Kenntnisse zu Projekt und Prozessmanagement, Verständnis zur Struktur von Managementsystemnormen	
Literaturempfehlungen	[1] R. von Rössing, Auditing Business Continuity: Global Best Practices. Rothstein. [2] British Standards Institution, “BS 25999 Business continuity,” British Standards Institution. [3] E. Swartz, D. Elliott, und B. Herbane, Business Continuity Management: A Crisis Management Approach. Routledge. [4] ISACA, CISA Review Manual. ISACA. [5] IT-Governance Institute, “CobiT,” IT Governance Institute. [6] European Network and information Security Agency (ENISA), “ENISA approach to Business Continuity for SMEs.” [7] Crisis Solutions, Exercising for Excellence: Delivering a Successful Business Continuity Management Exercise. BSI British Standards Institution. [8] International Organization for Standardization, “ISO/IEC 13335 series,” ISO. [9] International Organization for Standardization, “ISO/IEC 20000 series,” ISO. [10] International Organization for Standardization, “ISO/IEC 270xx series,” ISO. [11] Bundesamt für Sicherheit in der Informationstechnik (BSI), “IT-Grundschutz-Standards.” [12] OGC, ITIL v3 Core Publications. OGC. [13] CLUSIF, “MEHARI - Method.” [14] CERT, SEI, “OCTAVE (Operationally Critical Threat, Asset, and Vulnerability Evaluation).” [15] Österreichisches Normungsinstitut, “ONR 4900x series - ON-Regeln „Risikomanagement für Organisationen und Systeme“,” ON. [16] BKA und A-SIT Zentrum für sichere Informationstechnologie - Austria, “Österreichisches Informationssicherheitshandbuch.” [17] J. Burtles, Principles and Practice of Business Continuity. Rothstein. [18] J. Sharp, The Route Map to Business Continuity Management: Meeting the requirements of BS 25999. BSI British Standards Institution.	
Learning Outcomes	Comprehensive knowledge and understanding for preparing and taking decisions in a professional environment; heuristic competencies for leading roles as IS manager / IS officer in organisations; the ability to assert oneself in the management of organisations even beyond the scope of technical developments and necessities.	

Titel der Lehrveranstaltung	Security Management 1 – Basics and Systems
Umfang	2 SWS, 3 ECTS
Lage im Curriculum	2. Semester (VZ) bzw. 3. Semester (BB)
Lehr- und Lernformen	Arbeitsbasis ist eine Übungsfirma (Video, Unternehmensdokumentationen), darauf aufbauend werden in Gruppenarbeiten in Projektform die Themen erarbeitet, Quiz, Vortrag, Stationenbetrieb
Prüfungsmodalitäten	Schriftliche Prüfung
Learning Outcomes	▪ The students are familiar with basic structures of and considerations regarding management systems, especially in the context of information security and IT management. ▪ Through the supervised familiarisation with processes and the acquisition of project knowledge, the students are enabled to independently draw up process maps for security management systems, thereby synthesising their knowledge.
Course Contents	Among other things, the students learn the basics of management systems under ISO 27001, ISO 20000, ISO Annex SL, and CobIt. Fundamentals of project and process management in the context of strategic options for information security. Linked to this, the students develop basic project management knowledge and gather their first small project experiences through group work – always in the context of security management.
Titel der Lehrveranstaltung	Security Management 2 – Codes of Practice
Umfang	4 SWS, 4 ECTS
Lage im Curriculum	3. Semester (VZ) bzw. 4. Semester (BB)
Lehr- und Lernformen	Arbeitsbasis ist eine Übungsfirma (Video, Unternehmensdokumentationen), darauf aufbauend werden in Gruppenarbeiten in Projektform die Themen erarbeitet, Quiz, Vortrag, Stationenbetrieb
Prüfungsmodalitäten	Schriftliche Prüfung
Learning Outcomes	▪ The students are able to integrate norms, standards, guidelines, and regulations from the context of information security into the management systems from “Security Management 1 – Basics and Systems”. ▪ Thanks to their ability to carry out efficacy assessments, the students experience the success stories of their own work and are enabled to present these performances in conformity with the management later on as well. ▪ Based on their fundamental knowledge of IS-relevant norms and standards, the students are able to prepare and carry out according certifications.
Course Contents	▪ Norms, standards, guidelines, and regulations in the context of information security pursuant to ISO 27001, ISO 27002, and further guidelines of ISO 27k, ISO 15408, the BSI’s handbook on basic IT security, the Austrian handbook on IT security, the COBIT, etc. ▪ Fundamental knowledge of standardisation, the role of standardisation institutes ▪ Overview of essential IS-relevant norms and standards, legal basics of standardisation, accreditation, certification ▪ Building on the requirements for controlling information security in organisations of all sectors and sizes, situation-specific topic characteristics are addressed in smaller groups as well.

Titel der Lehrveranstaltung	IT Processes according to ITIL
Umfang	2 SWS, 3 ECTS
Lage im Curriculum	3. Semester (VZ+ BB)
Lehr- und Lernformen	Gruppenarbeiten in Projektform, Quiz, Vortrag, Stationenbetrieb
Prüfungsmodalitäten	Schriftliche Prüfung
Learning Outcomes	▪ The students put the basics of IT security management and ITIL frameworks into practice in an organisational environment and in an economically appropriate manner. ▪ The acquisition of the internationally renowned ITIL Foundation Certificate is perfectly suited to professional qualification. ▪ After completing this course, the students are also able to analyse and create IT service management processes.
Course Contents	The students acquire a sound knowledge of best practices in IT service management according to ITIL V3. In this context, the contents of the ITIL v3 Core publications on service strategy, service draft, service transition, service operation, and continuous service improvement are taught in particular. The course encompasses the preparation material for the internationally recognised ITIL Foundation Certificate. Upon request, the students can acquire the Foundation Certificate outside the degree programme at the end of the semester. Moreover, a typical product lifecycle, role description and casting (separation of powers as a security measure), and product approval process are described. In addition, topics such as ROI, risk management, and outsourcing within the framework of IT service management are also addressed. In the course of the five lifecycle phases, the following processes are examined: service strategy demand management, financial management, service portfolio management, service design, service level management, service catalogue management, supplier management, availability management, capacity management, IT security management, IT service continuity management, service transition, transition planning and support, change management, service asset and configuration management, release and deployment, service validation and test, evaluation management, service knowledge management, service operations, incident management, request fulfilment, event management, problem management, access management, continual service improvement, 7-step-improvement process, service measurement, service reporting, SLM-SIP (Service Improvement Plan)
Titel der Lehrveranstaltung	Security Management 3 – Risk and Emergency
Umfang	4 SWS, 5 ECTS
Lage im Curriculum	4. Semester (VZ) bzw. 5. Semester (BB)
Lehr- und Lernformen	Gruppenarbeiten in Projektform, Quiz, Vortrag, Stationenbetrieb
Prüfungsmodalitäten	Schriftliche Prüfung
Learning Outcomes	▪ The students acquire knowledge on the reasons and motives for developing a sustainable business continuity and risk management programme and thus attain the necessary competence for implementing emergency management systems and/or risk management methods in organisations or projects. ▪ By referencing the most important business continuity and risk management standards and processes, the students' work generates higher appreciation and efficiency.

	▪ Practical examples concerning the stages of the business continuity lifecycle (such as the implementation and evaluation of a business impact analysis, the development of strategies, and the business game) facilitate the transition from theory to entrepreneurial practice.

Course Contents	In this subject, the students accumulate their interdisciplinary competencies in project and process management already acquired in Security Management 1 and 2. Risk and emergency management are taught on the basis of projects and are carried out in a process-oriented manner. The interfaces and maintenance requirements often constitute critical success factors, as this course makes clear.

Operating Systems

Modulnummer:	Modultitel:	Umfang:
ITB1	Operating Systems	18 ECTS
Studiengang	Bachelor IT Security	
Lage im Curriculum	1.-3. Semester (VZ+ BB)	
Zuordnung zu den Teilgebieten	IT-Betrieb	
Niveaustufe	Basis	
Vorkenntnisse	Informatik: Geläufigkeit in der Bedienung der Hard- und Software von Personal Computern: Betriebssysteme mit grafischer Benutzeroberfläche, Office-Systeme (Textverarbeitung, Tabellenkalkulation). Vorausgesetzt wird das Modul „Netzwerke Grundlagen“ (NWT1). In den Betriebssystem-LVs werden auch Grundkenntnisse in der Programmierung (Modul Grundlagen) vorausgesetzt. Die Lehrveranstaltungen bauen inhaltlich aufeinander auf.	
Geblockt	Die einzelnen LVs werden geblockt abgehalten	
Kreis d. TeilnehmerInnen	Studierende des Studiengangs IT Security	
Beitrag zu nachfolgenden Modulen	Voraussetzung für Modul Betrieb vernetzter Systeme	
Literaturempfehlungen	[1] M. Kofler, Linux: Das umfassende Handbuch. Rheinwerk Computing. [2] J. Turnball, Hardening Linux. Apress [3] M.D. Bauer, Linux Server Security. O‘Reilley [4] M. Bishop, Computer Security: Art and Science. Addison-Wesley. [5] A.S. Tanenbaum, Modern Operating Systems, Prentice Hall. [6] J. McKellar u. A.Rubini u. J.Corbert u. G. Kroah-Hartman, Linux Device Drivers. O‘Reilley. [7] K. Brown, Programming Windows Security. Addison-Wesley.	
Learning Outcomes	▪ The students install and configure client and server systems in Windows and Linux and set up network services in such a way that security gaps are minimised or avoided entirely. ▪ The students manage company networks with central mechanisms (e.g., GPOs in AD networks). ▪ They configure auditing and logging for recording (critical) system processes. ▪ They use (shell) script languages for automating administration processes and carrying out analyses. ▪ The students apply various tools for hardening Windows and Linux operating systems. ▪ They are able to contrast Windows and Unix security concepts, related them to each other, and configure and secure a heterogeneous IT infrastructure.	
Titel der Lehrveranstaltung	Windows System Administration	
Umfang	3 SWS, 5 ECTS	
Lage im Curriculum	1. Semester (VZ+BB)	
Lehr- und Lernformen	ILV	

Prüfungsmodalitäten	LV-immanenter Prüfungscharakter im Übungsteil, praktische Arbeit (Adaptierung des Übungsnetzwerks) und Theoriefragen als Abschlussprüfung.
Learning Outcomes	▪ The students are able to explain the security concept of Windows. ▪ They instal, configure, and integrate Windows servers and workplace systems into the computer network. ▪ They set up local users and groups with their respective authorisations. ▪ They set up the Active Directory as the central user database and configuration solution. ▪ Using group guidelines, the students design security solutions for operating system and network environments in simulated company environments and instal the solutions in the network lab. ▪ At the end of the course, they can set up central services such as file or printing services and implement the necessary security settings
Course Contents	Windows server and client operating system installation and deployment, configuring local users and groups, (NTFS) authorisations, local system configuration; setting up AD DS, DNS, DHCP, configuration of work stations and security settings via GPOs.
Titel der Lehrveranstaltung	Unix System Administration
Umfang	3 SWS, 4 ECTS
Lage im Curriculum	2. Semester (VZ+BB)
Lehr- und Lernformen	ILV
Prüfungsmodalitäten	LV-immanenter Prüfungscharakter im Übungsteil, praktische Arbeit (Adaptierung des Übungsnetzwerks) und Theoriefragen als Abschlussprüfung.
Learning Outcomes	▪ The students install, configure, and integrate Linux systems in the computer network. ▪ They operate a Linux PC on a command line interface and understand and autonomously enter a little more complex shell commands. ▪ They are familiar with UNIX standard rights in detail. They protect resources from unauthorised/unwanted access through correctly allocated rights. ▪ They can move within the Unix file system tree, configure file systems on hard disk drives, make them available for use (locally and via NFS) and use them. ▪ At the end of the course, they are able to handle Unix processes. They use common text evaluation and manipulation tools to evaluate (logging) protocols.
Course Contents	The Unix file tree and the Unix standard rights, access control lists, configuration of local users and groups, configure hard disk drives and file systems, working with processes, confiture services (systemd), working on the command line (bash), text analysis and manipulation tools, IP network connection.
Titel der Lehrveranstaltung	Server Operation and Hardening Windows
Umfang	2 SWS, 3 ECTS
Lage im Curriculum	2. Semester (VZ) bzw. 3. Semester (BB)
Lehr- und Lernformen	ILV

Prüfungsmodalitäten	LV-immanenter Prüfungscharakter im Übungsteil, schriftliche (eCampus) Abschlussprüfung

Learning Outcomes	▪ The students can carry out security analyses and configurations using the tools included in Windows. ▪ They take measures to counter traditional threats. ▪ They apply (central) monitoring and configure (centrally controlled) updates. ▪ The students configure security baselines and analyse current logging information against these baselines. They use best practice analysis tools to assess the security configurations of typical server applications. ▪ The students assess the security characteristics of the authentication processes used in Windows. ▪ They set up security configurations for company networks using group guidelines in AD.
Course Contents	Analysis, logging, and auditing tools in Windows operating systems. Group guidelines as central instrument of a network-wide security setting. Authentication processes used by Windows. Setup and working with baselines. Update management.
Titel der Lehrveranstaltung	Server Operation and Hardening Unix
Umfang	4 SWS, 6 ECTS
Lage im Curriculum	2. Semester (VZ) bzw. 3. Semester (BB)
Lehr- und Lernformen	ILV
Prüfungsmodalitäten	LV-immanenter Prüfungscharakter im Übungsteil, praktische und schriftliche (eCampus) Abschlussprüfung
Learning Outcomes	▪ The students independently form their own kernel (e.g., by removing unnecessary modules to reduce threat exposure), prevent the loading of modules, use capabilities, and configure an LSM implementation like SELinux as a Mandatory Access Control solution. ▪ They set up a central user database (LDAP, AD) and configure Linux servers as clients. ▪ At the end of the course, the students are able to harden frequently used Linux services such as ssh, stunnel, sudo, etc., configure the authentication for various services (PAM), set up logging/auditing, and configure a Linux firewall. ▪ The students carry out security analyses using common tools in Linux. ▪ They configure Linux computers for various security requirements. ▪ They set up a heterogeneous IT network with Windows and Linux computers and compare their security concepts.
Course Contents	Linux kernel: modules and hardening of a kernel, use of capabilities, LSM implementation using the examples of SELinux and AppArmor with MAC. Authentication processes, name services and their installation or integration (also in a heterogeneous network), hardening Linux services (ssh, stunnel, sudo, etc.)

Secure Operation of Networked Systems

Modulnummer:	**Modultitel:**	**Umfang:**
ITB2	**Secure Operation of Networked Systems**	**21 ECTS**
Studiengang	Bachelor IT Security	
Lage im Curriculum	3., 4. und 6. Semester (VZ) bzw. 5.- 7. Semester (BB)	
Zuordnung zu den Teilgebieten	IT-Betrieb	
Niveaustufe	Anwendungen	
Vorkenntnisse	Technische Grundlagen (TECH1), Betriebssysteme (ITB1), Netzwerke Grundlagen (NWT1), Kryptografie (SITEC1) und Hardwaretoken Überfachliche Kompetenz „Leistungsfähigkeit“ Im Rahmen von Security Audits & Penetration Testing: ▪ Ressourcenmanagement: Umgang mit Stress, dessen Bewertung und Coping-Strategien ▪ Leistungsfähigkeit und Belastbarkeit Überfachliche Kompetenz „Selbstführung“ im Rahmen von Security Audits & Penetration Testing: ▪ Selbstmanagement und seine Teilkompetenzen: selbstständige Motivation, Zielsetzung, Planung, Zeitmanagement, Organisation, Lernfähigkeit, Erfolgskontrolle	
Geblockt	Die einzelnen LVs werden geblockt abgehalten	
Kreis d. TeilnehmerInnen	Studierende des Studiengangs IT Security	
Beitrag zu nachfolgenden Modulen		
Literaturempfehlungen	[1] F. Rommel, Active Directory Disaster Recovery. Packt Publishing. [2] R. W. Smith, Advanced Linux Networking. Addison-Wesley. [3] E. Marcus und H. Stern, Blueprints for High Availability: Designing Resilient Distributed Systems. John Wiley & Sons. [4] T. Clark, Designing Storage Area Networks. Addison-Wesley. [5] R. Cannings, H. Dwivedi, und Z. Lackey, Hacking Exposed Web 2.0: Web 2.0 Security Secrets and Solutions. McGraw-Hill Osborne. [6] M. Davis, S. Bodmer, und A. LeMasters, Hacking Exposed: Malware & Rootkits Secrets & Solutions. McGraw-Hill Osborne. [7] K. Schmidt, High Availability and Disaster Recovery: Concepts, Design, Implementation. Springer. [8] T. Gallagher, B. Jeffries, und L. Landauer, Hunting Security Bugs. Microsoft-Press. [9] I. C. Champ, L. Chaffin, und A. Chuvakin, Infosecurity 2008 Threat Analysis. Syngress Media. [10] J. De Clercq und G. Grillenmeier, Microsoft Windows Security Fundamentals. Digital Press. [11] A. S. Tanenbaum, Moderne Betriebssysteme. Pearson. [12] I. Alshanetsky, php|architect's Guide to PHP Security|. Marco Tabini & Associates, Inc. [13] T. Ballad und W. Ballad, Securing PHP Web Applications. Addison-Wesley. [14] A. Tiensivu, Securing Windows Server 2008: Prevent Attacks from Outside and Inside Your Organization. Syngress Media. [15] J. Clarke, SQL Injection Attacks and Defense. Syngress Media. [16] M. Cross, Web Application Security: A Guide for Developers and Penetration Testers. Syngress Media.	

	[17] R. Shimonski, Windows Server Clustering und Load Balancing. McGraw-Hill Osborne. [18] J. Seitz, Mehr Hacking mit Python. [19] D. Kennedy et. al., Metasploit: The Penetration Tester's Guide. [20] J. Erickson, The Art of Exploitation (2nd Edition). [21] J. Koziol et.al., The Shellcoder's Handbook. [22] D. Stuttard, M. Pinto, The Web Application Hacker's Handbook (2nd Edition).
Learning Outcomes	▪ Comprehensive knowledge of the workings of essential technical components for modern IT applications on the basis of web applications. ▪ Additionally, understanding of common security weaknesses in web applications as well as test methods and procedures for discovering them (penetration testing). ▪ Practical knowledge in dealing with security analysis tools and application case-specific selection of hacking tools.
Titel der Lehrveranstaltung	Database Systems
Umfang	3 SWS, 4 ECTS
Lage im Curriculum	4. Semester (VZ) bzw. 5. Semester (BB)
Lehr- und Lernformen	ILV, Vorlesung und praktische Übungen
Prüfungsmodalitäten	Übung: Abgaben von 2 Übungsbeispielen: 1.Bsp. SQL-Aufgaben (einzeln), 2.Bsp. Eine komplette Datenbank entwerfen, implementieren, befüllen; je eine Stored procedure und einen Trigger schreiben, BenutzerInnen mit verschiedenen Berechtigungen anlegen (einzeln) 2/3 Vorlesung: Schriftliche Abschlussprüfung 1/3
Learning Outcomes	▪ The students can explain the technological basics and architectures of database systems. (2) ▪ They are able to demonstrate the handling of data in database systems using SQL language. (3) ▪ The students can apply database technologies as subcomponents of other (web) applications. (3)
Course Contents	▪ Database draft: entity-relationship model, relational model, relation synthesis ▪ SQL: select, insert, update, delete, create, alter, drop, grant, revoke, deny, commit, rollback ▪ TransactSQL: syntax, cursor, stored procedures und functions, trigger ▪ Authorisation concept, database encryption ▪ Architecture of a database system: data system with query optimisation, access system with functioning of common index types, storage system with transaction management ▪ Tuning of a query ▪ Basic possibilities of accessing a database from a (web) application ▪ Basic structure of XML, XSD, and XSLT
Titel der Lehrveranstaltung	Essential Hacking Tools and Techniques

Umfang	3 SWS, 4 ECTS

Lage im Curriculum	4. Semester (VZ) bzw. 5. Semester (BB)
Lehr- und Lernformen	ILV, Vorlesung und praktische Übungen
Prüfungsmodalitäten	Beurteilung Präsentation von Tools und Techniken Beurteilung Protokoll Abschlussbericht einer CTF Simulation
Learning Outcomes	▪ The students understand the different stages of a professional hacking attack (e.g., Cyber Kill Chain). ▪ They can set up a professional hacking lab environment for practically testing and exercising attacks. ▪ The students are able to apply relevant hacking tools such as Nmap, Wireshark, Metasploit & Co. in practice. ▪ They can use Kali Linux as a professional platform for hackers and penetration testers. ▪ The students understand how exploits are carried out in order to compromise Windows and Linux systems and gain access to them. ▪ They understand the procedures and techniques that hackers use for attacking web servers and web applications such as, e.g., the workings of SQL Injection, Cross Site Scripting, etc. ▪ The students can apply the methods and tools for hacking WLAN and mobile systems.
Course Contents	▪ Setup of a suitable hacking lab ▪ Stages of a professional hacking attack ▪ Tools and methods used by attackers to find out all they need to know about their victims ▪ Optimal use of Nmap for analysing the target network’s vulnerabilities ▪ The hacking tool Wireshark ▪ Familiarisation with and application of Pen-test Distribution Kali Linux and its tool frameworks for the different attack phases ▪ Using the Metasploit framework for preparing and implementing diverse attacks ▪ Using techniques for protecting payloads against discovery by AV systems and IDS/IPS. ▪ Using privilege escalation to take the decisive step for completely taking over the victim system ▪ Tools for cracking passwords in different scenarios ▪ Getting to know how SQL Injection, XSS, and other attacks on web applications work ▪ What to do in case of WLAN hacking ▪ Cracking cryptographic algorithms such as WEP, WPA, and WPA2 ▪ Scenario-based application of techniques and tools
Titel der Lehrveranstaltung	Web and Application Security
Umfang	3 SWS, 5 ECTS
Lage im Curriculum	3. Semester (VZ) bzw. 6. Semester (BB)
Lehr- und Lernformen	ILV, Vorlesung und praktische Übungen
Prüfungsmodalitäten	Übung: Mitarbeit im Unterricht 50% Vorlesung: Klausur 50%

Learning Outcomes	▪ The students can explain the most common classes of vulnerabilities in web applications. (2) ▪ They can check an existing web application for classes of vulnerabilities and explain the vulnerability they have found. (3) ▪ The students are able to select and systematically apply the tools they need for checking vulnerabilities in web applications. (3)

Course Contents	Common classes of vulnerabilities in web applications (OWASP TOP10): ▪ A1 – Injection ▪ A2 – Broken Authentication and Session Management ▪ A3 – Cross-Site Scripting ▪ A4 – Broken Access Control ▪ A5 – Security Misconfiguration ▪ A6 – Sensitive Data Exposure ▪ A7 – Insufficient Attack Protection ▪ A8 – Cross-Site Request Forgery (CSRF) ▪ A9 – Using Components with Known Vulnerabilities ▪ A10 – Underprotected APIs
Titel der Lehrveranstaltung	Security Audits & Penetration Testing
Umfang	4 SWS, 6 ECTS
Lage im Curriculum	6. Semester (VZ) bzw. 7. Semester (BB)
Lehr- und Lernformen	ILV, Vorlesung und praktische Übungen
Prüfungsmodalitäten	Beurteilung Protokoll Permission to Attack Beurteilung Protokoll Abschlussbericht Penetration Test
Learning Outcomes	▪ The students understand the basic procedure of a penetration test. (2) ▪ They can organise and carry out a penetration test on their own. (3) ▪ The students are able to find technical vulnerabilities within the framework of a penetration test and explain them in a report. (3) ▪ During a penetration test, they can select and systematically implement the tools that they need for the verification of vulnerabilities. (3) ▪ The students apply their self-made Python programmes to a penetration test. ▪ Based on the test results, the students can issue recommendations for repairing or mitigating the findings.
Course Contents	▪ Security vulnerability research & disclosures ▪ Legal framework conditions for security research ▪ Common Vulnerability Scoring Subsystem (CVSS) ▪ Security testing standards and methodologies (OSSTMM, PTES, ISSAF, A7700) ▪ Penetration testing with Python ▪ Prototypical procedure and phases of penetration testing projects ▪ Independent organisation and implementation of a real penetration test in groups ▪ Penetration test reporting ▪ Listing mitigation or repair recommendations
Titel der Lehrveranstaltung	Business Continuity and Disaster Recovery

Umfang	2 SWS, 2 ECTS

Lage im Curriculum	6. Semester (VZ) bzw. 7. Semester (BB)
Lehr- und Lernformen	Übung
Prüfungsmodalitäten	Prüfungsimmanent, d.h. in der Übung wird der aktuelle Projektfortschritt täglich 1-2x überprüft 50% Abschließend ist eine umfassende Dokumentation des Gesamtprojekts einzureichen die das Projekt und dessen Ergebnisse darstellt. Diese wird durch eine Kurzpräsentation je Gruppe vorgestellt. 50%
Learning Outcomes	▪ The students can establish a defined system landscape. (3) ▪ They can analyse an existing system landscape with regard to its resilience. (4) ▪ The students are able to analyse the reasons for system failures and independently carry out compensating measures. (3) ▪ Based on the prescribed organisation scenario, the students can develop their ability to deal with conflict and achieve compromise. (3)
Course Contents	Practical exercise of emergencies in the field of business continuity and disaster recovery. To the extent possible under lab conditions, the students set up a computer centre operation with several companies and a fictitious Internet with service providers and draft a business continuity plan. Subsequently, they are confronted with a disaster situation within the framework of a business game and need to practically test the security measures contained in their plan with regard to their resilience.

Basics of Network Technology

Modulnummer:	Modultitel:	Umfang:
NWT 1	Basics of Network Technology	16 ECTS
Studiengang	Bachelor IT Security	
Lage im Curriculum	1. und 2. Semester (VZ+BB)	
Zuordnung zum Teilgebiet	Netzwerktechnik	
Niveaustufe	Einführung	
Vorkenntnisse	Keine	
Geblockt	Die einzelnen LVs werden geblockt abgehalten	
Kreis der TeilnehmerInnen	Studierende des Studienganges IT Security	
Beitrag zu nachfolgenden Modulen	Voraussetzung für Modul Network Security und Modul IT Betrieb 2	
Literaturempfehlungen	[1] R. Froom, B. Sivasubramanian, und E. Frahim, Building Cisco Multilayer Switched Networks (BCMSN). Cisco Press. [2] A. Leinwand, B. Pinsky, und M. Culpepper, Cisco Router Configuration. Cisco Press. [3] R. Desmeules, Cisco Self-Study: Implementing Cisco IPV6 Networks. Cisco Press. [4] J. F. Kurose und K. W. Ross, Computer Networking: A Top-Down Approach. Addison-Wesley. [5] A. S. Tanenbaum, Computer Networks. Prentice Hall. [6] D. E. Comer, Computer Networks and Internets with Internet Applications. Prentice Hall. [7] A. S. Tanenbaum und M. van Steen, Distributed Systems: Principles and Paradigms. Prentice Hall. [8] C. Spurgeon, Ethernet: The Definitive Guide. O'Reilly Media. [9] R. Perlman, Interconnections: Bridges, Routers, Switches, and Internetworking Protocols. Addison-Wesley. [10] D. E. Comer, Internetworking with TCP/IP, Vol 1. Prentice Hall. [11] P. Loshin, IPv6: Theory, Protocol, and Practice, 2nd Edition. Morgan Kaufmann. [12] E. Stein, Rechnernetze und Internet. Hanser. [13] J. Doyle und J. Carroll, Routing TCP/IP, Volume 1. Cisco Press. [14] J. Doyle und J. Carroll, Routing TCP/IP, Volume 2. Cisco Press. [15] W. R. Stevens, TCP/IP Illustrated I: The Protocols: 001. Addison-Wesley Longman Verlag.	
Learning Outcomes	▪ The students are able to assess the requirements for the IT infrastructure of local networks and they understand the interplay of the components used. ▪ They can analyse protocols in terms of their security and configure frequently used services (http, DNS/DNSSEC, etc.). ▪ They are able to create technical documentations, assess information offers, and correctly cite sources.	
Titel der Lehrveranstaltung	Introduction to Networks and Distributed Systems	
Umfang	5 SWS, 9 ECTS	
Lage im Curriculum	1. Semester (VZ+BB)	

Lehr- und Lernformen	Theorie: Präsentation incl. Fallbeispiele für IP Subnetting und Paketanalyse Praxis: Aufgabenstellungen die tw. im Sebststudium gelöst werden Laborübung: praktische Umsetzung der Inhalte aus der Vorlesung inklusive einer technischen Dokumentation.
Prüfungsmodalitäten	▪ Schriftliche Abschlussprüfung ▪ Immanente Überprüfung durch Übungsaufgaben ▪ Dokumentation der Laborübungen
Learning Outcomes	▪ The students are familiar with the basics of IP networks and can allocate the necessary functions to the individual OSI layers. ▪ They know the individual LAN components including their tasks (functions) and are able to manage them. ▪ In addition, they can develop small-scale company networks (max. 100 workplaces). ▪ The students are able to break down network protocols into their functions and display them accordingly (packet analysis) ▪ Students can explain text types and their differences. ▪ Students can create text types of the type documentation.
Course Contents	In this course, basic network concepts are discussed based on the ISO/OSI-7 Layers Model and the TCP/IP Model and these models are compared with others. For a better understanding, the terminology is demonstrated by presenting real examples (e.g., from Windows/Unix). The basic terms of network technology are introduced, e.g., OSI Model, Bus Network Topology/Star LAN /WAN, connection-oriented versus connectionless services, wired/wireless, broadcast networks/point-to-point/uni-/broad-/multicasting, Switch/Router/Gateway, etc. physical layer, electrotechnical basics of data transmission, cable norms, LWL, fibre optical cable, WLAN Ethernet: 802.3, CSMA/CD, norms for 10 Mbit- 10 Gbit IP: IPv4 and IPv6, subnetting and supernetting TCP: Sliding Windows; RTT calculation, network address translation (NAT), port address translation (PAS). First security considerations, denial of services. In this course, (simple) networks based on Ethernet and TCP/IP (with static routing) with Cisco components are set up and packet analyses are conducted. The practical teaching is based on Cisco components. In the course part relating to metadisciplinary competences (text type 1), the students get an overview of the following text types: documentations, reports incl. executive summaries, e-mails, guidelines. In addition, the students learn the basics of proof-reading texts. The focus of this course is on the text type documentation in network technology.
Titel der Lehrveranstaltung	Active Components and Application Protocols
Umfang	6 SWS, 7 ECTS
Lage im Curriculum	2. Semester (VZ+BB)
Lehr- und Lernformen	Theorie: Präsentation inkl. Fallbeispiele Laborübung: praktische Umsetzung der Inhalte aus der Vorlesung inklusive einer technischen Dokumentation
Prüfungsmodalitäten	Vorlesung: Abschließende Theorieprüfung Dokumentation der Laborübungen Praktische Abschlussübung

Learning Outcomes	▪ The students get to know a selection of common services (HTTP, DNS/DNSSEC, SMTP, POP3, IMAP, SSL, etc.) and learn how to configure them in Linux. ▪ The students are able to analyse the protocols with regard to security. ▪ The students are familiar with the basics of local routing and switching technologies. ▪ They can assess them according to set requirements and understand the interrelations of the various components in detail. ▪ The students can conduct advanced searches in scientific databases and evaluate the appropriateness of the information they have gathered. ▪ The students can cite correctly.
Course Contents	The following application protocols are discussed in detail: DNS: basic DNS architecture, DNS organisation, DNS data (resource records, names and zones, zone definition), BIND server configuration, DNSSEC: new resource records, “Path of Trust”, trust anchor, DNS cache poisoning, attack vectors HTTP: architecture of web applications, protocol differences 0.9./1.0/1.1/2.0, important HTTP – header e-mail: SMTP protocol, relaying, POP3, IMAP, e-mail source code (header interpretation), postfix configuration routing: *static/dynamic routing * distance vector protocols (RIP, EIGRP under IPv4 and IPv6)* LinkState protocols (OSPF) switching: *switching technologies and net design * VLAN/VTP* Ether channel * L2 security (port security, etc.) * redundant L2 systems (spanning tree) In this part of the course (metadisciplinary competence: handling information), students learn how to find, assess, and use information correctly. The contents include an introduction into subject-specific information resources (online resources of the library, scientific search engines, etc.), methods to analyse the quality of information and citation methods.

Network Security

Modulnummer:	Modultitel:	Umfang:
NWT 2	Network Security	10 ECTS
Studiengang	Bachelor IT Security	
Lage im Curriculum	3. und 4. Semester (VZ) bzw. 4. Semester (BB)	
Zuordnung zum Teilgebiet	Netzwerktechnik	
Niveaustufe	Anwendungen	
Vorkenntnisse	Module Netzwerke Grundlagen (NWT1), Kryptografie (SITEC1), Betriebssysteme (ITB1).	
Geblockt	Die einzelnen LVs werden geblockt abgehalten	
Kreis der TeilnehmerInnen	Studierende des Studiengangs IT Security	
Beitrag zu nachfolgenden Modulen	IT Betrieb 2	
Literaturempfehlungen	[1] M. Shema und B. Johnson, Anti-Hacker Tool Kit. McGraw-Hill Osborne. [2] K. Lam, D. LeBlanc, und B. Smith, Assessing Network Security. Microsoft-Press. [3] E. D. Zwicky, S. Cooper, und D. B. Chapman, Building Internet Firewalls. . [4] H. Benjamin, CCIE Security Exam Certification Guide. Cisco Press. [5] D. Gollmann, Computer Security. John Wiley & Sons. [6] O. Kyas und M. a Campo, IT Crackdown.Sicherheit im Internet. mitp. [7] C. Peikari und A. Chuvakin, Kenne deinen Feind. O'Reilly Media. [8] S. Convery, Network Security Architectures. Cisco Press. [9] W. Stallings, Network Security Essentials. Applications and Standards. . [10] C. Kaufman, R. Perlman, und M. Speciner, Network Security: Private Communication in a Public World. Prentice Hall. [11] W. Stallings, SNMP, SNMPv2, SNMPv3, and RMON 1 and 2. Addison-Wesley. [12] J. Canavan, The Fundamentals of Network Security. . [13] W. Böhmer, Virtual Private Networks (VPN). Hanser. ;	
Learning Outcomes	▪ The students are able to assess requirements for IT networks (LAN and WAN) and develop the solutions needed. ▪ They are familiar with the common infrastructure and know how to administer and manage it. ▪ The students are able to analyse existing systems in terms of their vulnerabilities (network design, access, etc.) and develop alternative solutions.	
Titel der Lehrveranstaltung	Network Security Components	
Umfang	2 SWS (LB) + 1 SWS (VO), 3 ECTS (LB) + 2 ECTS (VO)	
Lage im Curriculum	3. Semester (VZ) bzw. 4. Semester (BB)	
Lehr- und Lernformen	Theorie: Präsentation incl. Fallbeispiele Laborübung: praktische Umsetzung der Inhalte aus der Vorlesung inclusive einer technischen Dokumentation	

Prüfungsmodalitäten	Vorlesung: Abschließende Theorieprüfung Dokumentation der Laborübungen
Learning Outcomes	▪ The students can select security components such as firewalls, demilitarised zones, and VPN gateways for the respective requirements and integrate them into existing networks. ▪ They are able to configure and administer current products by leading manufacturers. ▪ They understand the concepts of various security protocols (IPSEC, etc.) and can categorise them according to cryptographic concepts, and apply them. ▪ The students are able to assess the security requirements of companies in terms of network protocols and implement them.
Course Contents	Fundamentals of firewalling and packet filtering (stateless filtering and stateful packet inspection) proxy, application-level gateway and traffic inspection (circuit-level proxies and application-level proxies), TLS interception or breaking up of encrypted traffics (deep packet inspection), VPN (virtual private networks), basics of tunnelling and VPN, IPSec and SSL-based VPN network design and planning with regard to a FW concept (DMZ design), checking firewall rulesets with PortScan
Anmerkungen	Die Lehrveranstaltung verwendet teilweise Basisinformationen aus den LVs „Aktive Komponenten", „Applikationsprotokolle und Protokollanalyse" sowie „Serverbetrieb und Hardening"
Titel der Lehrveranstaltung	Secure Networks
Umfang	4 SWS, 5 ECTS
Lage im Curriculum	4. Semester (VZ+ BB)
Lehr- und Lernformen	Theorie: Präsentation incl. Fallbeispiele Laborübung: praktische Umsetzung der Inhalte aus der Vorlesung inclusive einer technischen Dokumentation
Prüfungsmodalitäten	Dokumentation der Laborübung Umsetzung einer Fallstudie inclusive einer Verteidigung der Lösung
Learning Outcomes	▪ The students are able to plan up-to-date security technologies for radio networks and to carry out error analyses specifically for layers 1 and 2. ▪ They can develop security requirements for corporate networks (LAN and WLAN) and select and implement the relevant technologies. ▪ The students are able to implement and administer 802.1X and EAP solutions for both LAN and WLAN. ▪ They are familiar with AAA infrastructures (RADIUS, TACACS+) and can implement central authentication servers. ▪ They can carry out the above-mentioned points for a large number of end devices in an automated manner using Active Directory.
Course Contents	WLAN: fundamental mechanisms (congestion avoidance, virtual carrier sensing, 802.11n, 802.11ac, bridging, hotspot solutions, etc.), differences in security requirements compared to LAN authentication infrastructures and their practical implementation (RADIUS und TACACS+); protocol 802.1X and different EAP authentication forms (PEAP, EAP-TLS, EAP-FAST, etc.), integration of a company PKI and implementation in a typical Active Directory infrastructure.

Languages

Modulnummer:	**Modultitel:**	**Umfang:**
SPR1	**Languages**	**6 ECTS**
Studiengang	Bachelor IT Security	
Lage im Curriculum	1.- 3. Semester (VZ) bzw. 2.- 4. Semester (BB)	
Zuordnung zu den Teilgebieten	Sprachen	
Niveaustufe	B1-B2	
Vorkenntnisse	Maturaniveau	
Geblockt	Teilweise	
Kreis d. TeilnehmerInnen	Studierende des Studiengangs IT Security	
Beitrag zu nachfolgenden Modulen	Studierende erwerben die Fähigkeit LVs in englischer Sprache zu folgen und wissenschaftliche Arbeiten auf englisch zu verfassen. Weiters sind sie in der Lage effektive Präsentationstechniken in der Zielsprache anzuwenden und IT-bezogene Themen zu diskutieren.	
Literaturempfehlungen	Murphy, Raymond. *English Grammar in Use*. 2012: Cambridge UP. Cotton, David. *Market Leader*. 2010: Pearson Longman. Emmerson, Paul. *Business Grammar Builder* 2002: Macmillan. Taylor, Ken*. Business English*. 2006: Summertown Publishing. Skern, Tim. *Writing Scientific English*. 2011: Facultas. Macgilchrist, Felicitas. *Academic Writing*. 2014: Schoeningh. Goodson, Patricia. *Becoming an Academic Writer*. 2013: Sage. Allsop Jace and Tricia Aspinall. *BEC Vantage Testbuilder*. 2004: Macmillan	
Learning Outcomes	▪ Command of basic and complex English grammatical structures ▪ Oral and written English communication skills in a professional context ▪ Application of effective presentation techniques in the target language ▪ Skills for applying for relevant job openings in English ▪ Implementation of basic conventions of academic writing ▪ Acquisition of language command for the successful completion of international standardised assessment tests	
Titel der Lehrveranstaltung	**English 1**	
Umfang	2 SWS, 2 ECTS	
Lage im Curriculum	1. Semester (VZ) bzw. 2. Semester (BB)	
Lehr- und Lernformen	Unter Berücksichtigung der grundlegenden Fertigkeiten des Sprachenerwerbs (Schreiben, Sprechen, Lesen, Sprachverständnis) wird der Lehrstoff in didaktisch variierter Form vermittelt: Gruppenarbeiten, Einsatz von Videos, Flipped Classroom, Vorträge, praktische Anwendung der Theorie (z.B. Präsentationen, academic writing), etc.	
Prüfungsmodalitäten	Continuous Assessment und Abschlussprüfung	
Learning Outcomes	▪ The students gain an insight into the requirements of standardised “language assessment tests”. ▪ They acquire the skill to communicate orally and in writing in a professional context, to engage in job interviews, and to hold presentations. ▪ Through the use of various anglophone media, the students have access to current topics in English.	

Course Contents	Professional English: Business communication, career change, presentations, executive summary, current IT-related topics in anglophone media, BEC exam preparation, language structure

Titel der Lehrveranstaltung	**English 2**
Umfang	2 SWS, 2 ECTS
Lage im Curriculum	2. Semester (VZ) bzw. 3. Semester (BB)
Lehr- und Lernformen	Unter Berücksichtigung der grundlegenden Fertigkeiten des Sprachenerwerbs (Schreiben, Sprechen, Lesen, Sprachverständnis) wird der Lehrstoff in didaktisch variierter Form vermittelt: Gruppenarbeiten, Einsatz von Videos, Flipped Classroom, Vorträge, praktische Anwendung der Theorie (z.B. Präsentationen, academic writing).
Prüfungsmodalitäten	Bewertung der sprachlichen Qualität einer schriftlichen Arbeit, welche im Rahmen der LV "Sicherheitsmanagement 1 – Basics und Systeme" auf Englisch verfasst wird. 2. Continuous assessment 3. BEC Prüfung auf CEFR level B2
Learning Outcomes	▪ Based on the learnt theory on the topic of “writing a report” and its practical application within the framework of tasks in the course “Security Management 1 – Basics and Systems”, the students acquire the tools they need for writing subject-relevant texts.
Course Contents	1. Acquisition of interdisciplinary competencies – academic writing in an IT context in cooperation with a subject-relevant course (“BUS”). The focus of the students’ writing skills is on drawing up a “report” in English. 2. Further BEC exam preparation 3. Language structure
Titel der Lehrveranstaltung	**English 3**
Umfang	2 SWS, 2 ECTS
Lage im Curriculum	3. Semester (VZ) bzw. 4. Semester (BB)
Lehr- und Lernformen	Unter Berücksichtigung der grundlegenden Fertigkeiten des Sprachenerwerbs (Schreiben, Sprechen, Lesen, Sprachverständnis) wird der Lehrstoff in didaktisch variierter Form vermittelt: Gruppenarbeiten, Einsatz von Videos, Flipped Classroom, Vorträge, praktische Anwendung der Theorie (z.B. Präsentationen, academic writing), etc
Prüfungsmodalitäten	Continuous Assessment und Abschlussprüfung
Learning Outcomes	▪ The students are able to keep up with English-taught courses within the framework of the study programme. ▪ They acquired the ability to communicate in English on level B2 (oral and written). ▪ Having completed this course, the students are able to pass international standardised exams on level B2.
Course Contents	Further BEC exam preparation Language structure Application of relevant IT vocabulary in discussions and texts

Working Techniques

Modulnummer:	Modultitel:	Umfang:
ARB	**Working Techniques**	**23 ECTS**
Studiengang	Bachelor IT Security	
Lage im Curriculum	4. und 6. Semester (VZ) bzw. 5.- 7. Semester (BB)	
Zuordnung zu den Teilgebieten	Transferable Skills	
Niveaustufe	Anwender	
Vorkenntnisse	Keine	
Geblockt	Die einzelnen LVs werden geblockt abgehalten	
Kreis d. TeilnehmerInnen	Studierende des Studienganges IT Security	
Beitrag zu nachfolgenden Modulen		
Literaturempfehlungen	[1] K. Samac, M. Prenner, und H. Schwetz, Die Bachelorarbeit an Universität und Fachhochschule: Ein Lehr- und Lernbuch zur Gestaltung wissenschaftlicher Arbeiten. UTB. [2] M. Karmasin und R. Ribing, Gestaltung einer wissenschaftlichen Abschlussarbei. UTB. [3] R. Burkart, Kommunikationswissenschaft. UTB. [4] P. Atteslander, Methoden der empirischen Sozialforschung. Schmidt. [5] A. Beiderwieden und E. Pürling, Projektmanagement für IT-Berufe. Bildungsverlag Eins. [6] B. Jenny, Projektmanagement: Das Wissen für den Profi. Vdf Hochschulverlag. [7] A. F. Chalmers, N. Bergemann, und C. Altstötter-Gleich, Wege der Wissenschaft: Einführung in die Wissenschaftstheorie. Springer. [8] U. Eco, Wie man eine wissenschaftliche Abschlußarbeit schreibt. UTB. [9] H. Balzert, C. Schäfer, M. Schröder, und U. Kern, Wissenschaftliches Arbeiten - Wissenschaft, Quellen, Artefakte, Organisation, Präsentation. W3l. [10] J. A. Schülein und S. Reitze, Wissenschaftstheorie für Einsteiger. UTB.	
Learning Outcomes	▪ Target group-oriented creativity in formulation is fundamental when communicating information security content to specific target groups. ▪ The students are thus enabled to convey these situations to decision-makers in organisations and to document and implement the outcomes and outputs of these decision-making processes in a factual context. ▪ The students can present content in a situation-adequate and recipient-oriented manner, recognise the difference between everyday language, educational language, and specialist language, and implement it accordingly.	
Titel der Lehrveranstaltung	Methods of Scientific Writing	
Umfang	2 SWS, 4 ECTS	
Lage im Curriculum	4. Semester (VZ) bzw. 5. Semester (BB)	
Lehr- und Lernformen	Inverted Classroom	
Prüfungsmodalitäten	Bewertung der Abgabe	

Learning Outcomes	▪ The students successfully complete their scientific theses.
Course Contents	Module 1: Theoretical basics: scientific methods, planning and structuring of a scientific paper, concept, research problem and object, formal criteria of a scientific final thesis including citing, writing style, text structure, research, searching scientific databases, source requirements, methods of a scientific final thesis Module 2: Practical implementation: Within the framework of the course, the students write a seminar paper on a focus topic. In this paper, the state of the art in science and the research problems are elaborated. Both the content and the “formal” criteria are taken into account in the assessment. Interdisciplinary competence: Scientific work ▪ Types of research and research methods ▪ Questions of ethics in research ▪ Structure of a bachelor thesis ▪ Structuring research problems ▪ Planning a bachelor thesis ▪ Developing a research question ▪ Literature review/research ▪ Analysing scientific papers (recognising different types)
Titel der Lehrveranstaltung	Elective
Umfang	3 SWS, 7 ECTS
Lage im Curriculum	6. Semester (VZ+BB)
Lehr- und Lernformen	ILV
Prüfungsmodalitäten	immanenter Prüfungscharakter
Learning Outcomes	▪ The students acquire competencies in their chosen specialisation.
Course Contents	The objective of the specialisation is to promote students with regard to their chosen focus area. The courses offered cover advanced, in-depth knowledge in the field of IT security (e.g., honeypots, malware analysis).
Titel der Lehrveranstaltung	Bachelor Seminar with Bachelor Thesis
Umfang	3 SWS, 12 ECTS
Lage im Curriculum	6. Semester (VZ) bzw. 7. Semester (BB)
Lehr- und Lernformen	Inverted Classroom
Prüfungsmodalitäten	Vortrag und Abgabe der Arbeit

Learning Outcomes	Interdisciplinary competence: “Self-management”: ▪ The students are able to pursue targeted time management in order to flexibly work on diverse tasks within a given period. (3) ▪ They can hold short presentations on a complex topic.
Course Contents	Supervision of the bachelor thesis Interdisciplinary competence: Presenting 2: The part “Presenting 2” serves the purpose of specialisation in the topic “Speaking with Confidence”. In this context, the students deepen the following skills: ▪ Planning and carrying out bachelor presentations (5 minutes / 15 minutes) ▪ Reinforcing content with graphics / slides Interdisciplinary competence: Self-management: ▪ Self-management and its partial competencies: independent motivation, goal setting, planning, time management, organisation, learning ability, success monitoring ▪ Flexibility ▪ Basics of resilience

Vocational Internship

Modulnummer:	**Modultitel:**	**Umfang:**
ARB	**Vocational Internship**	**24 ECTS**
Studiengang	Bachelor IT Security	
Lage im Curriculum	5. Semester (VZ) bzw. 3.-7. Semester (BB)	
Zuordnung zu den Teilgebieten	Berufspraktikum	
Niveaustufe	Fortgeschrittene	
Vorkenntnisse	Fortgeschrittene Kenntnisse der IT Security als Voraussetzung für das Berufspraktikum	
Geblockt	Berufsbegleitend: nein, Vollzeit: ja	
Kreis d. TeilnehmerInnen	Studierende des Studienganges IT Security	
Beitrag zu nachfolgenden Modulen	Praxiserfahrung	
Literaturempfehlungen		
Learning Outcomes	▪ The students are able to present the professional field that they familiarised themselves with during their internship. (2) ▪ They can critically reflect on their acquired competencies in light of their internship experiences. (4) ▪ The students can derive fields of action for their personal, specialist, and professional further development from their internship experiences. (6)	
Titel der Lehrveranstaltung	Supervisory Seminar for the Internship	
Umfang	2 SWS, 4 ECTS	
Lage im Curriculum	5. Semester (VZ) bzw. 7. Semester (BB)	
Lehr- und Lernformen	Workshop-Charakter	

Prüfungsmodalitäten	Abgaben und Endpräsentation

Learning Outcomes	▪ The students are able to present the professional field that they familiarised themselves with during their internship. (2) ▪ They can critically reflect on their acquired competencies in light of their internship experiences. (4) ▪ The students can derive fields of action for their personal, specialist, and professional further development from their internship experiences. (6)
Course Contents	The students reflect on their internship experiences: ▪ Before the internship (expectations, objectives, course contents that are important for the internship) ▪ During the internship (areas of responsibility, comparison to expectations, comparison to course contents from the study programme) ▪ After the internship (collected experiences, derived personal fields of action)